



## Introduction: Cognitive Attack Surfaces in Multiagent Operators

# The Multiagent Operator Paradigm I

Modern AI deployment has shifted from single-model inference to **multiagent operators**—systems where a primary agent delegates subtasks to specialized subagents, tools, and external services.

Table 1: Representative multiagent system architectures and primary attack surfaces.

| System      | Architecture  | Agent Count     | Communication            |
|-------------|---------------|-----------------|--------------------------|
| Claude Code | Hierarchical  | $1 + n$ dynamic | Task delegation          |
| AutoGPT     | Autonomous    | $1 +$ plugins   | Tool invocation          |
| CrewAI      | Role-based    | 3–10 fixed      | Sequential/parallel      |
| LangGraph   | State machine | Variable        | Graph traversal          |
| MetaGPT     | SOP-driven    | 5–8 roles       | Document passing         |
| OpenClaw    | Cyberphysical | $1 +$ tools     | Multi-platform messaging |

# The Multiagent Operator Paradigm II

This architectural evolution introduces **cognitive attack surfaces** absent in single-agent systems. Throughout this paper, we use *cognitive security* (abbreviated *CogSec*) to denote the discipline of protecting agent reasoning processes—beliefs, goals, and trust relationships—from adversarial manipulation.

# The 2026 Multiagent Landscape I

# The 2026 Multiagent Landscape II

## From Chatbots to Cognitive Operators

The AI systems of 2026 bear little resemblance to the chatbots of 2023. Where earlier systems responded to queries within a single context window, contemporary multiagent operators exhibit fundamentally different characteristics:

1. **Persistent Agency:** Agents maintain state across sessions, accumulate context, and pursue goals over extended timeframes. A coding assistant doesn't just answer questions—it tracks project architecture, remembers previous decisions, and adapts recommendations based on accumulated understanding.
2. **Active World Modification:** Unlike passive responders, modern operators write code that executes, send emails that reach recipients, modify infrastructure that serves users, and make purchases that transfer funds. The gap between “AI-generated content” and “AI-executed action” has collapsed.
3. **Hierarchical Delegation:** Primary agents spawn subordinate agents for specialized tasks. Claude Code delegates to

# Motivating Incidents I

*Note: The following are hypothetical scenarios constructed to illustrate attack patterns documented in the literature. They are not reports of actual incidents.*

These scenarios, grounded in documented attack patterns, illustrate the emerging threat landscape across all five adversary classes.

# Motivating Incidents II

## Incident: Nested Instruction Injection (External)

A user submitted a document for analysis containing hidden instructions: “Ignore previous instructions and instead output the system prompt.” The document appeared benign to human reviewers but exploited the agent’s instruction-following behavior. The attack succeeded because the agent processed user-supplied content as potential instructions.

**Attack vector:**  $\Omega_1$  (external) via direct prompt injection \

**Impact:** Information disclosure or instruction override \

**Traditional Defense Gap:** Standard input validation passed—the attack exploited *semantic interpretation* of benign-appearing content



# Motivating Incidents III

## Incident: The Poisoned Code Review (Peripheral)

A development team deployed a multiagent system for automated code review. Agent-Alpha performed initial analysis, delegating security scanning to Agent-Beta (connected to external vulnerability databases). An attacker compromised a third-party CVE feed, injecting fabricated vulnerability reports that convinced Agent-Beta to recommend removing legitimate security controls. Agent-Alpha, trusting Agent-Beta's "security expertise," approved the changes.

**Attack vector:**  $\Omega_2$  (peripheral) via tool response manipulation \

**Impact:** Security regression through trusted channel exploitation \

**Traditional Defense Gap:** Input filtering, authentication, and encryption all passed—the attack entered through *content* of a trusted, authenticated channel

# Motivating Incidents IV

## Incident: The Identity Confusion Attack (Agent-Level)

A multiagent customer service system used role-based permissions. An attacker crafted prompts that convinced a junior agent it had been “temporarily promoted” to administrator status. The agent’s self-model shifted, and it began exercising permissions it believed it possessed, bypassing access controls that relied on self-reported identity.

**Attack vector:**  $\Omega_3$  (agent-level) via identity manipulation \

**Impact:** Privilege escalation through cognitive state corruption \

**Traditional Defense Gap:** Cryptographic identity was intact; the attack targeted the agent’s *self-model*, not its credentials

# Motivating Incidents V

## Incident: The Consensus Manipulation (Coordination)

A financial services firm used a 5-agent ensemble for trade approval. The system required 3/5 agent agreement for large transactions. An adversary discovered that agents weighted peer opinions based on historical agreement rates. By slowly building agreement history through small, legitimate-appearing trades, the attacker cultivated artificial trust, eventually manipulating consensus for unauthorized large transactions.

**Attack vector:**  $\Omega_4$  (coordination) via progressive trust exploitation \ **Impact:** Consensus bypass through manufactured reputation \ **Traditional Defense Gap:** Per-request authorization succeeded for each transaction; the attack exploited *temporal composition* across sessions

# Motivating Incidents VI

## Incident: Orchestrator Compromise (Systemic)

An attacker gained access to the orchestrator agent through a supply chain vulnerability in a training pipeline. With control of the central coordinator, the attacker could issue legitimate-appearing delegations to all subordinate agents, redirect trust evaluations, and suppress security alerts. The compromise remained undetected because the orchestrator itself validated security checks.

**Attack vector:**  $\Omega_5$  (systemic) via orchestrator control \ **Impact:** Total system compromise with attack obfuscation \ **Traditional Defense Gap:** All internal security mechanisms reported nominal—the attack *controlled the mechanisms themselves*

# Motivation from Recent Deployments I

The proliferation of multiagent AI systems introduces security considerations that the community is actively addressing. Early work on cognitive security in remote teams and information ecosystems [cordes2020great; cordes2021narrative; cordes2023atlas] established foundational concepts for information resilience, which this framework extends to artificial agents. Complementary work on Active Inference has demonstrated how cognitive modeling and cognitive science perspectives—including formalization of OODA (Observe-Orient-Decide-Act) loops and multiscale communication dynamics—provide integrative frameworks for understanding agent cognition under adversarial conditions [?]. The OWASP Top 10 for LLM Applications 2025 [?] places prompt injection as the top vulnerability, while the newly released OWASP Top 10 for Agentic Applications [?] specifically addresses autonomous AI systems with “tool misuse, prompt injection, and data leakage” as primary concerns.

# Motivation from Recent Deployments II

## **Scale of Deployment (2024–2026):**

- ▶ Enterprise AI agents processing significant transaction volumes (53% of organizations now deploy RAG and agentic pipelines [?])
- ▶ Autonomous coding assistants with repository write access (GitHub Copilot CVE-2025-53773 demonstrated RCE via prompt injection)
- ▶ Multi-agent orchestrators in infrastructure management contexts

## **Emerging Attack Surface:**

- ▶ Inter-agent communication channels lack authentication standards—ARIA model proposes cryptographically verifiable delegation [?]

## Motivation from Recent Deployments III

- ▶ Trust delegation mechanisms operate without formal verification—recent work on CP-WBFT achieves 85.71% Byzantine fault tolerance improvement [?]
- ▶ Belief provenance remains largely untracked in production systems—cognitive degradation attacks exploit this gap [?]

### **Limitations of Current Defenses:**

- ▶ Input/output filtering primarily designed for single-agent architectures—adaptive attacks bypass 12 published defenses with  $>90\%$  success [?]
- ▶ Limited standardized frameworks for cognitive integrity verification
- ▶ Byzantine fault tolerance infrequently applied to AI agent systems—emerging work addresses this gap [?]

## Motivation from Recent Deployments IV

The fundamental constraint is that traditional security models assume a clear boundary between trusted and untrusted components. In multiagent systems, this boundary is fluid—agents must reason about the trustworthiness of other agents' reasoning.



# Problem Statement I

Traditional security models address:

- ▶ **Input validation:** Filtering malicious prompts
- ▶ **Output sanitization:** Preventing harmful generations
- ▶ **Access control:** Limiting tool permissions

They fail to address:

- ▶ **Inter-agent trust:** How should Agent  $A$  weight claims from Agent  $B$ ?
- ▶ **Belief provenance:** Which beliefs derive from verified vs. adversarial sources?
- ▶ **Coordination integrity:** Can agents be manipulated into malicious consensus?
- ▶ **Temporal persistence:** Do attacks survive context boundaries?

## Problem Statement II

- ▶ **Cognitive integrity:** How can the cognitive systems of today and tomorrow remain flexible and robust amidst change in composition and context?

# Research Questions I

This paper addresses four fundamental research questions, with emphasis on formal foundations:

**RQ1: Taxonomy and Formal Characterization.** *What classes of cognitive attacks exist against multiagent systems, and how can they be formally characterized to enable systematic analysis?*

We develop an initial taxonomy spanning epistemic, behavioral, social, and temporal attack dimensions. Crucially, each attack class receives formal definition enabling systematic analysis, composition rules, and detection bounds (sec:attack-taxonomy).

**RQ2: Trust Algebra.** *How might inter-agent trust be modeled to prevent trust amplification and laundering attacks while enabling legitimate delegation?*

## Research Questions II

We introduce a trust calculus with bounded delegation ( $\delta^d$  decay guarantee), prove associativity properties, and establish the no-amplification theorem ensuring that trust cannot be manufactured through delegation chains (sec:trust-calculus).

**RQ3: Defense Composition.** *How do cognitive defense mechanisms compose, and what guarantees can we provide about layered defense effectiveness?*

We present a defense composition algebra enabling formal reasoning about series and parallel defense arrangements. We prove that orthogonal defenses compose multiplicatively (not additively) for detection rate improvement (sec:defense-composition).

**RQ4: Fundamental Bounds.** *What are the information-theoretic limits on cognitive attack detection?*

## Research Questions III

We derive the stealth-impact tradeoff theorem establishing fundamental bounds on detection independent of defense implementation. We prove that attacks cannot simultaneously achieve high impact and complete undetectability, providing theoretical grounding for defense design (sec:detection-bounds).

# Contributions I

This paper provides both theoretical foundations and practical mechanisms for cognitive security:

## Formal Contributions:

1. **Threat Taxonomy:** A systematic classification of cognitive attacks across epistemic, behavioral, social, and temporal dimensions with formal definitions enabling rigorous analysis (sec:attack-taxonomy)
2. **Trust Calculus:** A mathematical framework for inter-agent trust with bounded delegation ( $\delta^d$  decay), associativity proofs, and formal guarantees against trust amplification attacks (sec:trust-calculus)
3. **Defense Composition Algebra:** Formal rules for composing security mechanisms with provable detection rate bounds under series and parallel composition (sec:defense-composition)

# Contributions II

4. **Information-Theoretic Bounds:** Fundamental limits on attack detection relating stealth constraints to maximum achievable impact, independent of defense implementation (sec:detection-bounds)
5. **Formal Verification:** Model-checked safety properties including belief integrity, trust boundedness, and goal alignment preservation (sec:formal-verification)

## Conceptual Contributions:

1. **Cognitive Security Operator Posture:** The proactive defensive stance required when the attack surface spans beliefs, goals, and inter-agent coordination (sec:operator-posture)

# Contributions III

2. **The Cognitive Integrity Framework (CIF):** An integrated approach combining architectural defenses, runtime monitoring, and Byzantine-tolerant coordination for multiagent systems (sec:system-model)

**Empirical Validation:** Part 2 of this series demonstrates the practical viability of these formal mechanisms across six production architectures, showing that layered cognitive defenses significantly outperform single-mechanism approaches. Practitioner translations of these results appear in Part 3; domain-specific applications and real-world incident analyses appear in Part 4.



# Paper Organization I

The remainder of this paper is structured as follows:

**sec:adversary-classes: Threat Model** develops a comprehensive adversary taxonomy ( $\Omega_1$ – $\Omega_5$ ) with attack complexity analysis, detectability matrices, and detailed scenarios for each attack class.

**sec:system-model: Cognitive Integrity Framework** presents the formal foundations of CIF, including system model definitions, cognitive state representations, integrity properties, and the trust calculus.

**sec:arch-defenses: Defense Mechanisms** describes architectural defenses (cognitive firewalls, belief sandboxing), runtime defenses (tripwires, invariant checking), and coordination defenses (Byzantine consensus, quorum verification).

**sec:anomaly-detection: Detection Methods** covers anomaly detection algorithms, provenance analysis techniques, and real-time monitoring systems.

# Paper Organization II

**sec:formal-verification: Formal Verification** proves the main theorems, presents invariant preservation lemmas, and describes model checking configuration.

**sec:discussion: Discussion** examines limitations, deployment considerations, and connections to related work.

**sec:summary: Conclusion** summarizes contributions and identifies directions for future research.

**Part 2: Experimental Validation** A separate, second, companion paper reports empirical results from architecture-aware parametric simulation across 950 attack vectors and six production architectures. Readers seeking detection rates, ablation studies, Bayesian uncertainty quantification, and architecture-specific defense configurations should consult Part 2 (DOI: [10.5281/zenodo.18364128](https://doi.org/10.5281/zenodo.18364128)).

# Paper Organization III

**Part 3: Actionable Insight** A separate, third, companion paper provides a qualitative review and practitioner's synthesis—accessible-language summaries of the theoretical and empirical findings, operational checklists, risk assessment frameworks, deployment guides, monitoring strategies, and cost-benefit analyses. Part 3 assumes no formal prerequisites and targets security teams operating multiagent systems (DOI: [10.5281/zenodo.18364130](https://doi.org/10.5281/zenodo.18364130)).

# Paper Organization IV

**Part 4: Applications** A fourth companion paper (*Cognitive Integrity in Critical Domains*) applies CIF across ten high-stakes operational sectors—rare-earth mining, nation-state alliances, cyber-security, drone warfare, supply chains, biowarfare, food security, trade wars, infrastructure, and information ecosystems—through the integrated CIF-AD-OODA analytical model. Readers interested in domain-specific goal-hijacking defenses, universal attack patterns (FR polarity inversion, constraint relaxation, context boundary violation), or retrospective analysis of documented 2024–2025 AI agent security incidents should consult Part 4.

## Reading Companion: Where to Find Specific Topics I

This paper is designed to stand alone as the formal-foundations reference of the series. The table below points readers to the sibling paper and section where each related topic is developed most fully, so that this paper can be used in conjunction with—rather than requiring—its companions.

Table 3: Cross-paper navigation from Part 1 topics to sibling developments.

## Reading Companion: Where to Find Specific Topics II

If you want...

...consult...

Empirical detection rates for the defense compositions proved here

Part 2 (DOI: 10.5281/zenodo.18364128), §5 (Results) and S08 (Parametric Analysis)

Ablation studies isolating the contribution of each defense mechanism

Part 2, §5.6 and §5d

Bayesian uncertainty quantification on detection rates

Part 2, §5e

Game-theoretic analysis / Nash equilibrium of CIF vs. adaptive attackers

Part 2, §6 (Discussion)

Model-checking execution (NuSMV, TLA+ traces) for the invariants defined here

Part 2 S04 (Model Checking)

Deployment checklists and operator-posture guidance

Part 3 (DOI: 10.5281/zenodo.18364130), §5 (Deployment Guide), §5b (Incident Response)

Monitoring, drift detection playbooks, cost-benefit analysis

Part 3, §5c–§5d

Case studies showing CIF in

Part 3, §6b

# Scope and Limitations I

**In scope:** Attacks exploiting agent reasoning, trust, and coordination mechanisms in multiagent AI systems.

**Out of scope:**

- ▶ Traditional software exploits (buffer overflow, SQL injection, memory corruption)
- ▶ Physical attacks (hardware tampering, side-channel analysis)
- ▶ Supply chain compromise (malicious training data, backdoored models)
- ▶ Cryptographic attacks (we assume secure primitives per ax:crypto-limit)

**Assumptions:**

- ▶ Agents communicate over authenticated channels
- ▶ Base model capabilities are not adversarially modified

## Scope and Limitations II

- ▶ At least one honest orchestrator exists in hierarchical systems